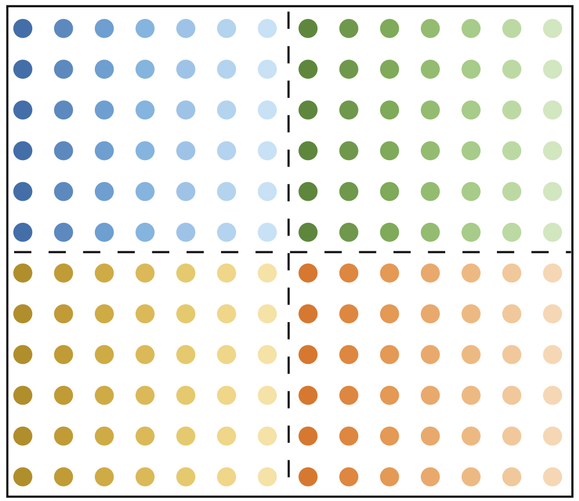
(a) Shifted-window boundary tokens

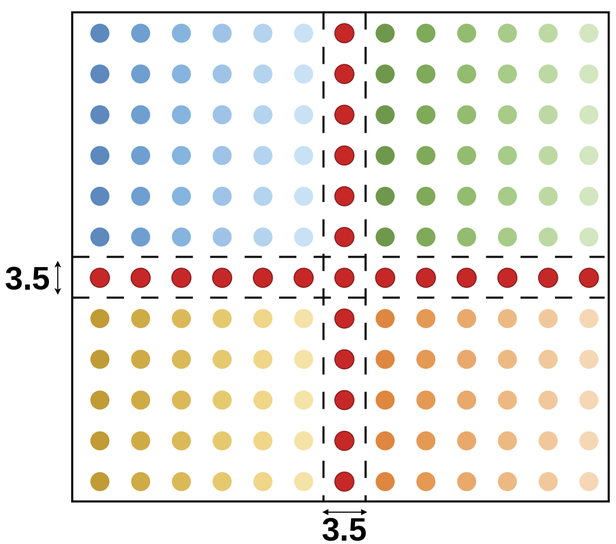
Toy shifted-window example (2×2 windows)

Regular partition



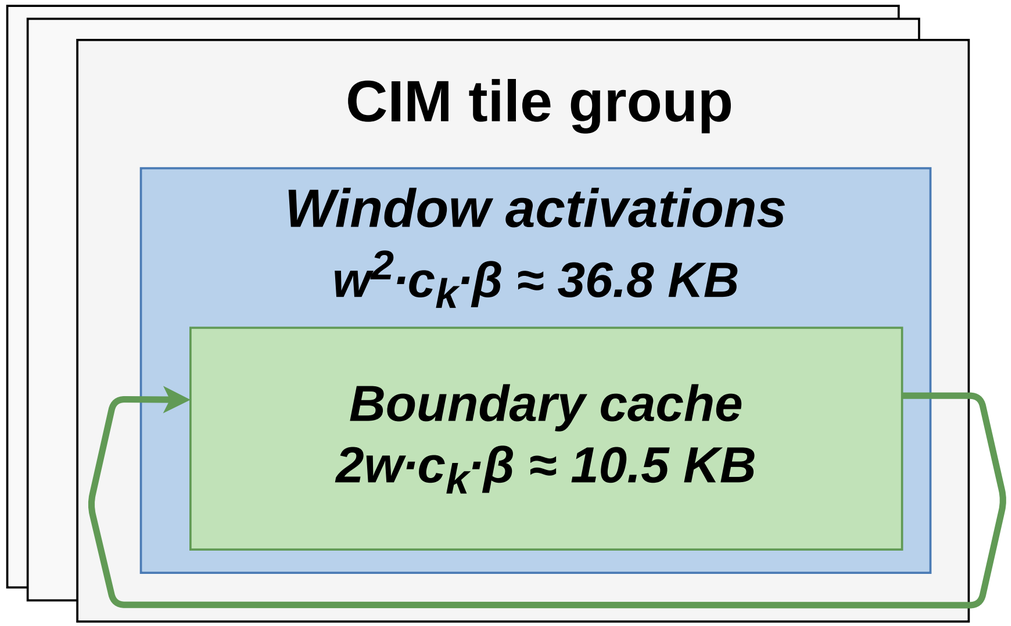
Shift creates reusable boundary tokens

Shifted partition



Without cache, these tokens are re-fetched from DRAM

(b) Boundary-cache layout



Spill-free target: $B_{min} = w(w+2) \cdot c_k \cdot \beta \approx 48$ KB

(c) Spill reduction @128 KB

Reuse boundary tokens; avoid DRAM re-fetch

Without boundary cache: 48.7% spill
With boundary cache: 4.5% spill
10.8× lower spill